

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-1876
Project Name	AI BlogNest API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current AI BlogNest API solution can be extended to support larger user bases, increased data loads, and additional features. It also records the roadmap for enhancing the backend with frontend support, advanced AI capabilities, search, user interaction, administration, personalization, and cloud deployment.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The current project is primarily a backend API and does not include a user-friendly frontend application.	Users need an API client to interact with the system directly.	Medium
2	Advanced blog search, filtering, categories and tags are not currently included.	Managing and discovering a larger collection of blogs can be less organized.	Medium
3	Comments, likes, user profiles, admin dashboard and personalized recommendations are not currently implemented.	User interaction, administration and personalization are limited in the current system.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	The backend provides authentication and protected API access using JWT.	Develop a user-friendly frontend application and extend the backend as the user base grows.
2	Data Storage	MongoDB stores user and blog information through Mongoose models.	Extend database organization with categories, tags, search and filtering for larger blog collections.
3	Performance	The application uses a modular Express.js architecture with separated routes, controllers and services.	Extend the modular architecture with additional services and cloud deployment as the system grows.
4	Security	JWT authentication and bcrypt password hashing are used for protected operations and password security.	Continue strengthening authentication and add administrative controls through the planned admin dashboard.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User-friendly frontend application and improved blog interaction.	Next Development Phase	Makes the API functionality easier for users to access and use.
Phase 2	AI-based blog title generation; AI-based grammar and content improvement; AI-generated images.	Future Enhancement	Expands AI-assisted content creation capabilities.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 3	Advanced blog search and filtering; categories and tags; comments and likes; user profile management.	Future Enhancement	Improves blog organization and user interaction.
Phase 4	Admin dashboard; personalized AI-based blog recommendations; cloud deployment; additional AI-powered content management.	Future Enhancement	Extends administration, personalization, scalability and deployment capabilities.